9.5 R-Combination

Let n and r be nonnegative integers with $r \leq n$. An **r-combination** of a set of n elements is a subset of r of the n elements. As indicated in Section 5.1, the symbol

$$\binom{n}{r}$$

which is read n choose r, denotes the number of subsets of size r (r-combinations) that can be chosen from a set of n elements.

9.5.1 Definition

The number of subsets of size r (or r-combinations) that can be chosen from a set of n elements, n, is given by the formula

$$\binom{n}{r} = \frac{P(n,r)}{r!}$$

or, equivalently,

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

where n and r are nonnegative integers with $r \leq n$.

9.5.2 Permutations with sets of Indistinguishable Objects

Suppose a collection consists of n objects of which

n_1 are of type 1 and are indistinguishable from each other. n_2 are of type 2 and are indistinguishable from each other. $\vdots$ n_k are of type k and are indistinguishable from each other.

and suppose that $n_1+n_2+\cdots+n_k = n$. Then the number of distinguishable.

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots\binom{n-n_1-n_2-\cdots-n_kk-1}{n_k}$$

$$\frac{n!}{n_1!n_2!n_3!\cdots n_k!} \tag{1}$$

9.6 r-Combinations with Repetition Allowed

An **r-combination with repetition allowed**, or **multiset of size *r***, chosen from a set X of n elements is an unordered selection of elements taken from X with repetition allowed. If $X = \{x_1, x_2, \ldots, x_n\}$, we write an r-combination with repetition allowed,or multiset of size r, as $[x_{i1}, x_{i2}, \ldots, x_i]$, where each x_ij is in X and some of the x_ij may equal.

9.6.1 Theorem

The number of r-combinations with repetition allowed (multisets of size r) that can be selected from a set of n elements is

$$\binom{r+n-1}{r}$$

This equals the number of ways r objects can be selected from n categories of objects with repetition allowed.